Evaluation Methods in Text Summarization

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As text summarization models evolve from early extractive methods to highly fluent, abstractive Large Language Models (LLMs), the reliable, automated evaluation of their outputs remains a critical bottleneck. Traditional lexical overlap metrics, such as ROUGE, provide computationally efficient baselines but heavily penalize valid paraphrasing and fundamentally fail to detect factual inconsistencies. To address the paraphrase penalty, the field transitioned toward semantic evaluation metrics like BERTScore, which leverage contextual word embeddings to assess underlying meaning rather than strict vocabulary. However, the persistent vulnerability of these reference-based metrics to fluent hallucinations necessitated a further paradigm shift toward reference-free, Natural Language Inference (NLI) frameworks, such as SummaC, to guarantee factual entailment. By synthesizing the theoretical mechanics of these methodologies and reviewing existing SummEval benchmark data, this report illustrates that evaluating modern summarization systems requires a composite approach: combining semantic metrics for fluency with NLI-based verification for factual integrity.

I. Introduction

Text summarization is a foundational task in Natural Language Processing (NLP) aimed at distilling the most important information from a source document into a concise and fluent text. While summarization models have advanced from early extractive methods to modern generative Large Language Models (LLMs), the reliable evaluation of these summaries remains a significant bottleneck. Evaluating a summary is a highly subjective problem that requires assessing fluency, coherence, relevance, and factual consistency. Human evaluation has been the gold standard, however it is time-consuming, and difficult to scale, so the history of summarization evaluation is defined by the development of automatic metrics that aim to correlate with human judgment.

The development of standardized automatic evaluation began in the early 2000s with the introduction of lexical overlap metrics, most notably ROUGE (Recall-Oriented Understudy for Gisting Evaluation). ROUGE revolutionized the field by providing a fast, deterministic way to evaluate generated summaries by counting n-gram matches against human-written reference summaries. For over a decade, ROUGE was the undisputed standard, functioning exceptionally well for extractive summaries.

However, the transition to the deep learning era and abstractive summarization, where models can generate entirely new sentences and complex paraphrases, exposed critical flaws in traditional lexical metrics. Because ROUGE penalizes valid synonyms, the field evolved toward semantic evaluation with metrics like BERTScore, which utilize contextual embeddings to measure meaning rather than strict word-matching. However a new critical failure was also introduced: "hallucinations" or the generation of factually incorrect information. Because both ROUGE and BERTScore primarily measure similarity to a human reference, they often fail to penalize these subtle factual errors. This necessitated a further shift towards newer evaluation methods like SummaC, which frame evaluation as a Natural Language Inference (NLI) task to ensure the generated summary logically follows the original source text.

II. ROUGE Evaluation Method

The historical standard for evaluating text summarization is **ROUGE** (Recall-Oriented Understudy for Gisting Evaluation), introduced by Lin 2004 [1]. ROUGE assesses the quality of a generated candidate summary by comparing it to a human-written reference summarie. This metric was historically favored, and remains highly effective, for evaluating *extractive* summarization. Because extractive models generate summaries by copying verbatim words and sentences directly from the source document, evaluating them inherently relies on strict lexical matching. If an extractive system successfully identifies and extracts the most relevant sentences, its output will naturally share a high degree of exact word overlap with a human reference summary.

A. N-grams and the ROUGE Formula

To measure this overlap, ROUGE relies on the concept of **n-grams**. An n-gram is defined as a contiguous sequence of n items (in this context, words) from a given text. For example, a unigram ($n = 1$) evaluates individual words, while a bigram ($n = 2$) evaluates pairs of consecutive words, allowing for a basic evaluation of word order.

At its core, ROUGE calculates recall. For a given n-gram length N, the ROUGE-N score is computed by counting the number of matching n-grams between the generated summary and the reference summary, divided by the total number of n-grams in the reference summary. The formal definition is given by the following equation:

$$ROUGE\text{-}N = \frac{\sum_{S\in\{Reference\}} \sum_{gram_n \in S} Count_{match}(gram_n)}{\sum_{S\in\{Reference\}} \sum_{gram_n \in S} Count(gram_n)} \tag{1}$$

Where $Count_{match}(gram_n)$ represents the maximum number of n-grams occurring in both the generated candidate summary and the human reference summary, ensuring the metric captures how much of the human reference was reproduced.

B. Variants of ROUGE

ROUGE constitutes a family of evaluation measures rather than a single mathematical check. The most prevalent variants utilized in modern NLP literature include:

- **ROUGE-1 and ROUGE-2:** These measure unigram and bigram overlap, respectively. ROUGE-1 is generally considered a strong proxy for capturing the core informational content and isolated keywords, while ROUGE-2 captures a greater degree of fluency and phrasing by enforcing word-pair matches.

- **ROUGE-L:** This variant utilizes the Longest Common Subsequence (LCS). Unlike strictly contiguous n-grams, LCS identifies the longest sequence of words that appear in both the generated and reference summaries in the same relative order, even if they are interrupted by other text. This naturally captures broader sentence-level structure without requiring predefined n-gram lengths.

C. Evolution: ROUGE 2.0

While standard ROUGE has served as the baseline for nearly two decades, its strict reliance on exact string matching limits its utility when evaluating abstractive models capable of paraphrasing. To address these rigidities, Ganesan 2018 [2] introduced **ROUGE 2.0**. This updated framework improves upon the original metric by introducing synonym dictionaries and advanced stemming techniques. Consequently, ROUGE 2.0 allows for the matching of semantically equivalent words (e.g. scoring a match between "huge" and "gigantic"), offering a vital bridge between strict lexical overlap and semantic evaluation, while simultaneously optimizing the evaluation pipeline for multi-document summarization tasks.

D. Limitations of ROUGE

Despite its ubiquity and computational efficiency, ROUGE possesses several critical limitations that render it insufficient for evaluating modern, advanced NLP systems, particularly those built on generative Large Language Models (LLMs). These limitations primarily rise from its foundational reliance on surface-level lexical matching:

- **The Paraphrase Penalty:** Standard ROUGE strictly measures syntactic overlap. Consequently, it severely penalizes abstractive summarization models that generate highly fluent summaries using synonyms or structural paraphrasing. For instance, if a reference summary contains the phrase “the automobile accelerated rapidly,” and the candidate summary generates “the car sped up quickly,” standard ROUGE yields a score of zero, failing to recognize the semantic equivalence.
- **Semantic Blindness:** Lexical overlap does not guarantee semantic preservation. Two sentences can share a high percentage of n-grams while conveying entirely different, or even opposite, meanings. For example, the addition or omission of a single negation word (e.g. “is” versus “is not”) drastically alters the semantic truth of a summary but results in only a negligible penalty in a ROUGE calculation.
- **Insensitivity to Factual Inconsistency:** Perhaps the most severe limitation in the context of modern generative AI is ROUGE’s inability to detect “hallucinations.” If an abstractive model generates a highly fluent summary that matches the syntactic structure and vocabulary of the reference text but alters a critical information, such as changing dates, names or a statistical figure, ROUGE may still output a deceptively high score. The metric lacks

the reasoning capability to verify if the generated facts are logically accurate to the source document.

These fundamental failure modes, the inability to measure underlying meaning and the failure to detect factual inconsistencies, necessitated a paradigm shift in summarization evaluation. To address the paraphrase penalty, the NLP development turned to contextual word embeddings (e.g. BERTScore), and to address factual hallucinations, the field moved toward Natural Language Inference frameworks (e.g. SummaC).

III. BERT Evaluation Method

As the limitations of lexical overlap became apparent with the rise of highly abstractive summarization models, the NLP community recognized the need to evaluate summaries based on their underlying semantic meaning rather than their surface-level syntax. This paradigm shift was heavily catalyzed by the advent of Transformer-based language models, most notably **BERT** (Bidirectional Encoder Representations from Transformers), introduced by Devlin et al. 2019 [3].

A. Contextual Embeddings as a Foundation

Unlike traditional word representations that assign a single, static vector to a word regardless of its usage, BERT generates *contextual embeddings*. This means the mathematical representation of a word dynamically changes based on the surrounding words in the sentence. Consequently, BERT captures complex semantic nuances and grammatical structure, providing a vector space where words with similar contextual meanings are close to one another.

B. BERTScore: Evaluating via Cosine Similarity

Leveraging the power of these pre-trained embeddings, Zhang et al. 2019 [4] introduced **BERTScore**, an automatic evaluation metric that fundamentally resolves the paraphrase issue in traditional metrics. Instead of relying on rigid n-gram overlaps, BERTScore evaluates a candidate summary by computing the semantic similarity between its tokens and the tokens of a reference summary.

Operating at a high conceptual level, the BERTScore pipeline avoids strict string matching. Instead, it computes the pairwise cosine similarity between the contextual embeddings of every token in the generated candidate summary and every token in the human reference. It then employs a greedy matching algorithm to align each candidate token to the most semantically similar reference token. These similarity scores are subsequently aggregated to compute precision, recall, and F1 measures.

C. Synthesis: Overcoming the Paraphrase Penalty

The integration of BERTScore represents a crucial evolution in summarization evaluation. Returning to the limitation examples established previously: if a reference summary contains “the automobile accelerated rapidly” and the candidate model generates “the car sped up quickly,” ROUGE heavily penalizes the generation. BERTScore, however, recognizes through its contextual embeddings that “automobile” and “car” occupy highly similar positions in the semantic vector space.

By successfully rewarding models for valid semantic preservation, BERTScore proves to be a vastly superior metric for evaluating the fluency and accuracy of modern abstractive systems. However, because BERTScore still relies fundamentally on a human reference summary, it assumes the reference itself is factual and complete. It is not explicitly designed to cross-reference the original source document, meaning it can still struggle to penalize subtle, highly fluent factual hallucinations, creating the need for the latest evolutionary step towards Natural Language Inference frameworks.

IV. SummaC Evaluation Method

While semantic metrics like BERTScore successfully addressed the paraphrase penalty, the adoption of generative Large Language Models (LLMs) exposed a new, more dangerous failure mode: hallucination. Highly parameterized abstractive models frequently generate text that is exceptionally fluent and grammatically perfect, but factually incorrect compared to the source document.

The negative impact of these hallucinations cannot be overstated. In high-stakes domains such as medical record summarization or legal document review, a synthesized summary that invents a dosage, alters a date, or flips a logical negation can lead to catastrophic real-world consequences. Because traditional metrics (ROUGE) and semantic metrics (BERTScore) primarily measure similarity against a human reference summary, they assume the generated text is factual as long as the vocabulary or embeddings match the reference. They do not cross-examine the original source text. Consequently, a highly fluent hallucination can achieve a deceptively high evaluation score.

A. Natural Language Inference (NLI) as Evaluation

To combat this, the evaluation paradigm required a shift from *reference-based* similarity to *reference-free* factual verification. The predominant method for achieving this is framing evaluation as a **Natural Language Inference (NLI)** task.

NLI is a foundational NLP classification task that takes two sentences, the“Premise” and the “Hypothesis” and determines their logical relationship. The model outputs a probability distribution across three classes:

- **Entailment:** The premise logically guarantees the truth of the hypothesis.
- **Contradiction:** The premise proves the hypothesis is false.
- **Neutral:** The premise does not provide enough information to prove or disprove the hypothesis.

In the context of summarization evaluation, the original source document acts as the Premise, and the generated summary acts as the Hypothesis. If a candidate summary is highly entailed by the source document, it is factually consistent. If it yields a high contradiction or neutral score, it contains hallucinations or unsupported additions.

B. Technical Architecture of SummaC

Applying standard NLI models directly to long documents is computationally expensive and often exceeds the token limits of Transformer models like RoBERTa. To solve this, Laban et al. 2022 [5] introduced **SummaC** (Summary Consistency), a framework that adapts NLI for document-level factual evaluation through a rigorous chunking and aggregation pipeline.

The SummaC architecture operates in three distinct phases:

1. Granular Chunking: Let the source document D be tokenized and split into m distinct sentences (or paragraphs), $S = \{s_1, s_2, \ldots, s_m\}$. Similarly, let the generated candidate summary Y be split into n distinct sentences, $Y = \{y_1, y_2, \ldots, y_n\}$.

2. Pairwise NLI Matrix Computation: An off-the-shelf NLI model (e.g., RoBERTa-large-MNLI) processes every possible pair of source sentences and summary sentences. For each pair (s_i, y_j), the model computes the probability of entailment. This generates an $m \times n$ entailment matrix E, where each element $E_{i,j}$ represents the entailment score of the i-th source sentence with respect to the j-th summary sentence:

$$E_{i,j} = P_{NLI}(\text{Entailment} \mid \text{Premise} = s_i, \text{Hypothesis} = y_j) \tag{2}$$

3. Aggregation: To calculate the final consistency score of the entire summary, SummaC aggregates this matrix. In its foundational Zero-Shot variant (SummaC$_{ZS}$), the framework assumes that for a summary sentence to be factual, it only needs to be fully entailed by at least *one* sentence in the source document. Therefore, it takes the maximum entailment score across the source document axis for each summary sentence, and then computes the mean of these maximums to output a global summary score:

$$\text{Score}(D, Y) = \frac{1}{n} \sum_{j=1}^{n} \max_{1 \le i \le m} (E_{i,j}) \tag{3}$$

By breaking the document into granular components and mathematically tracking the maximum logical entailment for every generated claim, SummaC provides a highly interpretable and robust defense against hallucinations, operating entirely independently of a human reference.

V. Meta-Evaluation: Comparing Metrics with SummEval

With a multitude of automated evaluation metrics available, ranging from lexical to semantic to factual, the NLP community requires a standardized method to evaluate the evaluators. This process, known as *meta-evaluation*, assesses how strongly an automated metric correlates with human judgment using statistical measures such as Pearson or Spearman correlation coefficients.

To facilitate this, Fabbri et al. 2021 [6] introduced **SummEval**, a comprehensive benchmark dataset designed specifically for the meta-evaluation of summarization systems. SummEval provides a vast corpus of generated summaries evaluated by human experts across four distinct quality dimensions: *Coherence*, *Consistency* (Factuality), *Fluency*, and *Relevance*.

Analyzing the performance of ROUGE, BERTScore, and SummaC through the lens of SummEval provides a definitive synthesis of their respective strengths and weaknesses:

- **Relevance and Fluency:** SummEval demonstrates that **BERTScore** significantly outperforms standard **ROUGE** in correlating with human judgments of fluency and relevance. Because BERTScore leverages contextual embeddings, it effectively rewards abstractive models that successfully capture the core meaning of the source text using diverse vocabulary, whereas ROUGE unfairly penalizes them.
- **Consistency (Hallucination Detection):** On the dimension of factual consistency, both ROUGE and BERTScore exhibit weak correlations with human annotators, confirming their vulnerability to fluent hallucinations. **SummaC**, however, achieves state-of-the-art correlation on this dimension. By framing evaluation as an NLI task directly against the source document, SummaC reliably aligns with human annotators in detecting unsupported additions and factual errors.

VI. Conclusion

The reliable evaluation of text summarization is a dynamic challenge that has evolved alongside advancements in text generation. This report synthesized this evolutionary trajectory by examining three pivotal methodologies: ROUGE, BERTScore, and SummaC.

Historically, the strict lexical overlap measured by ROUGE provided a necessary and computationally efficient baseline for extractive systems. However, the transition to highly fluent, abstractive Large Language Models exposed critical limitations in syntax-based evaluation. The rigid paraphrase penalty necessitated the adoption of semantic metrics like BERTScore, which successfully evaluate underlying meaning rather than strict vocabulary. Yet, the persistent threat of fluent, reference-matching hallucinations required a further paradigm shift toward factual verification, a problem effectively addressed by SummaC through reference-free Natural Language Inference.

Ultimately, the current landscape of NLP research demonstrates that no single automated metric is perfect. The robust evaluation of modern summarization systems inherently requires a composite approach: deploying semantic metrics like BERTScore to assess fluency and meaning preservation, while critically relying on NLI frameworks like SummaC to guarantee factual integrity.

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